

Device Tree Creation, Boot Script & SD Card Copy Procedure

DE10-Standard SoC FPGA (Cyclone V) — Technical Practical Guide (v2.1)

STEP 1: Create the Device Tree & Boot Script

Run the following commands in the Embedded Command Shell (SoC EDS) environment to convert the system configuration files into binary files for Linux and U-Boot:

1. Convert the HPS system design (`sopcinfo`) into Device Tree Source (`.dts`)

```
sopc2dts -i ../../RTL/soc_system/soc_system.sopcinfo -o socfpga.dts
-b ../../Quartus/soc_system_board_info.xml
-b ../../Quartus/hps_common_board_info.xml
--bridge-removal all --clocks
```

2. Compile the Device Tree Source (`.dts`) into Device Tree Blob (`.dtb`)

```
dtc -I dts -O dtb -o soc_system.dtb socfpga.dts
```

3. Compile the U-Boot boot script (`.scr`)

```
mkimage -A arm -O linux -T script -C none -a 0 -e 0 -n "Boot Script"
-d boot.script u-boot.scr
```

Important Note About the Two Board Description XML Files (`-b` Parameter)

The two files `soc_system_board_info.xml` and `hps_common_board_info.xml` passed through the `-b` option are extracted from the sample project directory on the Terasic SystemCD at:

```
C:\intelFPGA\de10\DE10-Standard_v.1.4.0_SystemCD\Demonstration\SoC_FPGA
```

You can obtain these two files from sample project directories such as `DE10_Standard_FB` or `DE10_Standard_GHRD`, depending on which sample configuration you choose as your reference.

Additional note: If the system reports that the `mkimage` tool is missing, install it using the package manager of the shell environment.

STEP 2: Copy the Files to the SD Card (Boot Preparation)

After the files have been successfully generated, copy the hardware configuration files and bootloader to the SD card so that the DE10-Standard board can boot.

Procedure:

1. Insert the SD card (with the Linux Console image already flashed) into the computer.
2. Open the SD card drive.

Note: Copy the files to the FAT32 partition displayed by the operating system.

1. Copy and overwrite the following three files in the root directory of the SD card:
2. `soc_system.dtb`: The Device Tree file generated in Step 1.
3. `u-boot.scr`: The U-Boot boot script compiled in Step 1.
4. `soc_system.rbf`: The Raw Binary file containing the FPGA hardware configuration (converted from the `.sof` file).
5. Safely eject the SD card from the computer, insert it into the SD card slot on the DE10-Standard board, and power on the board to begin the boot process.